Practice Test 4

Decision Making, Learning, the LMU, and Analysing Representation

Time: 30 minutes **Total: 20 marks (+2 bonus)**

Closed book. No notes, calculator, phone, or other devices.

Write on separate paper. Show your reasoning; partial credit is given for correct reasoning. Label sketch axes and important values. Exact or approximate values that need no calculator are sufficient. This is a practice assessment, not a prediction of the real test.

Question 1 [4 marks]

An accumulate-to-threshold decision model uses a 1-D LIF ensemble with radius 1. A recurrent connection computes the identity through a first-order synapse with $\tau = 0.1$ s. A constant evidence input u enters the ensemble with transform 1. Treat the input synapse as matched to the recurrent one, and let $x(0) = 0$.

(a) [1] Find A and B in the dynamics $\dot{x} = Ax + Bu$ that this network implements, and write the ideal $x(t)$ for constant u.

(b) [2] On one set of axes, sketch the decoded value $x(t)$ for $0 \le t \le 2$ s and $u = 0.2, 0.1, -0.1, -0.2$. Mark when the ideal solution reaches ± 1. Then explain why the simulated value stops changing near ± 1 instead of continuing to grow.

(c) [1] A second 1-D population receives choice(x), which is $+1$ if $x > 0.9$, -1 if $x < -0.9$, and 0 otherwise. Its neurons have encoders ± 1 and intercepts drawn uniformly from $[0.4, 0.9]$. Explain why its decoded output is *exactly* zero before a decision is reached, rather than only approximately zero.

Question 2 [4 marks]

The Legendre Memory Unit (LMU) implements $\dot{\mathbf{x}} = A\mathbf{x} + Bu$ with $\mathbf{x} \in \mathbb{R}^q$, window length θ, and, for $i, j = 0, \ldots, q-1$,

$$\theta A_{ij} = (2i+1) \begin{cases} -1, & i < j, \\ (-1)^{i-j+1}, & i \ge j, \end{cases} \qquad \theta B_i = (2i+1)(-1)^i.$$

(a) [1.5] For $q = 1$, write the LMU's differential equation. Describe in words what $x(t)$ represents about the recent input. Then give the recurrent and input transforms A' and B' that implement it with matched synapses, $\tau = 0.1$ s and $\theta = 0.5$ s.

(b) [1] A $q = 6$, $\theta = 0.5$ s LMU receives a 1 Hz sine for $0 \le t < 2$ s and a 2 Hz sine for $2 \le t < 4$ s. A category ($+1$ for 1 Hz, -1 for 2 Hz) is decoded from the LMU ensemble. Explain why the category cannot be decoded from the current input value $u(t)$ alone, but can be decoded from the LMU state.

(c) [1.5] With 1000 neurons and fixed θ and τ, q is swept over $1, 2, 4, 8, 16, 32, 64$. Sketch the expected classification RMSE against q (log axis). Give one reason for the behaviour at small q and one at large q.

Question 3 **[4 marks]**

A population with activities $a_j(x)$ produces the decoded output $\hat{y} = \sum_j d_j a_j(x)$. The decoders should be learned online so that $\hat{y}(t)$ tracks a desired signal $y^d(t)$.

(a) [2] Starting from the instantaneous loss $E = \frac{1}{2}\left(\hat{y} - y^d\right)^2$, derive the gradient-descent decoder update with learning rate η. A scalar post-population has gains α_i, encoders $e_i = \pm 1$, and weights $w_{ij} = \alpha_i e_i d_j$. Write the corresponding weight update, and state what plays the role of a modulatory input to post-neuron i.

(b) [1] The error population computes $\varepsilon = \hat{y} - y^d$. Predict what happens if this sign is reversed, with the learning rule unchanged, and explain why.

(c) [1] The network learns the identity from 2 Hz band-limited noise with RMS 0.3. The output RMSE is computed in five consecutive 2-s chunks of a 10-s run and averaged over runs. On one set of axes, sketch the chunk RMSE for $\eta = 10^{-4}$ and for $\eta = 10^{-5}$. Give one reason why the error does not reach zero.

Question 4 **[4 marks]**

A pendulum has angle θ measured from horizontal (counter-clockwise positive) and dynamics

$$mL^2\ddot{\theta} = \tau - mgL\cos\theta, \qquad m = L = 1, \quad g = 9.8 \text{ m/s}^2.$$

A PD controller applies $\tau_{PD} = k_p(\theta^* - \theta) - k_d\dot{\theta}$ with $\theta^* = 0$, $k_p = 49$, $k_d = 10$, and $\theta(0) = \pi/4$.

(a) [1] Estimate the steady-state angle under PD control alone. In which direction is it offset from the target?

(b) [1.5] An ensemble representing the pendulum state is connected to the torque through a PES connection initialized to compute zero. Its error signal is derived from the PD output. What function does this connection learn? Why is the PD output a suitable error signal although nothing supplies the correct torque? With $\varepsilon = \hat{y} - y^d$ as in Question 3, what is the sign of the PD output in the error?

(c) [1.5] On one set of axes, sketch $\theta(t)$ for $0 \le t \le 20$ s under PD control alone and with the adaptive ensemble. Label the initial angle, target and steady-state offset.

Question 5 **[4 marks]**

A population of N neurons represents a scalar $x \in [-1, 1]$.

(a) [1] To analyse which functions of x this population can decode, which matrix is decomposed with the SVD? Why is that the right matrix, rather than the decoders for one particular function?

(b) [1] For 100 LIF neurons with heterogeneous intercepts, sketch the singular value spectrum on a log axis. Add the noise floor for neuron noise variance σ^2, and indicate which components are usable.

(c) [2] In the rotated basis, the estimate is $\hat{x} = \sum_m \chi_m(x)\Phi_m$, where $\langle \chi_m \chi_n \rangle_x = S_m \delta_{mn}$. With independent neuron noise of variance σ^2, the expected error is

$$E = \left\langle \left(x - \sum_m \chi_m \Phi_m\right)^2 \right\rangle_x + \sigma^2 \sum_m \Phi_m^2.$$

Let $c_m = \langle \chi_m x \rangle_x$. Show that including mode m with its noise-free coefficient $\Phi_m = c_m/S_m$, rather than omitting it, changes the error by

$$\Delta E_m = \frac{c_m^2}{S_m}\left(\frac{\sigma^2}{S_m} - 1\right).$$

What does this imply about which modes, and therefore which functions, are usable?

Bonus **[2 marks]**

With the same learning rate and population, PES learns $f(x) = x$ much faster than $f(x) = x^2$. Averaged over inputs, PES performs gradient descent on the mean squared decoding error. Use the SVD analysis of the population to explain the difference in learning speed.

End – 20 marks + 2 bonus